

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0510 Slot A

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5) Assessment Tool Weightage: 40%

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A
Industry Problem Solving Task

Code of Conduct:

I M . Manoj Kumar(192425114) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M. Manoj Kumar

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Industry Problem	20%	Comprehensive understanding of real world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection

A city surveillance system must operate under the following conditions:

- If latency > 100 ms → system fails
- If accuracy < 85% → unacceptable
- Hardware supports only moderate computation

Models available:

Viola-Jones: 78%, low latency

YOLO: 92%, moderate latency

Deep Learning: 95%, high latency

Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

Ans:

The city surveillance system has three strict constraints: latency must be 100 ms or less, accuracy must be at least 85%, and the hardware can support only moderate computation. Therefore, a model must satisfy all three conditions simultaneously to be selected.

Viola-Jones provides low latency, so it satisfies the latency requirement. However, its accuracy is only 78%, which is below the required minimum of 85%. Therefore, it fails the accuracy constraint and cannot be selected, even though its computational requirement and latency are favorable.

Deep Learning provides the highest accuracy of 95%, so it satisfies the accuracy requirement. However, it has high latency and requires high computation. Since the system fails when latency exceeds 100 ms and the available hardware supports only moderate computation, Deep Learning does not satisfy the deployment constraints. Therefore, its higher accuracy cannot compensate for its inability to meet the latency and hardware requirements.

YOLO provides 92% accuracy and has moderate latency. Its accuracy is greater than the required 85%, and its moderate computational requirement matches the available hardware. Since its latency is described as moderate, it is the only model among the three that can reasonably satisfy the latency constraint while also meeting the accuracy and hardware requirements.

The comparison can therefore be expressed logically:

- Viola-Jones → Accuracy = 78% < 85% → Fails
- YOLO → Accuracy = 92% ≥ 85%, moderate computation and latency → Satisfies
- Deep Learning → Accuracy = 95% ≥ 85%, but high latency/computation → Fails deployment

constraints

Thus, YOLO should be selected for the city surveillance system. It provides a suitable balance between accuracy, latency, and computational requirements. The decision is not based only on which model has the highest accuracy; instead, all constraints must be satisfied. YOLO meets the require

2. Autonomous Vehicle Detection Logic

A vehicle detection system must:

- Prioritize safety (accuracy > 90%)
- Maintain response time < 50 ms

Options:

Method A: 88% accuracy, 30 ms

Method B: 93% accuracy, 70 ms

Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

Ans:

The autonomous vehicle detection system has two critical constraints. First, safety must be prioritized, so the accuracy must be greater than 90%. Second, the response time must be less than 50 ms. A suitable method must satisfy both conditions simultaneously.

Method A has an accuracy of 88% and a response time of 30 ms. Its response time satisfies the requirement because 30 ms is less than 50 ms. However, its accuracy is only 88%, which is below the required value of greater than 90%. Since safety is the primary requirement, Method A cannot be selected. Its fast response is not sufficient to compensate for inadequate detection accuracy.

Method B has an accuracy of 93%, which satisfies the safety requirement because it is greater than 90%. However, its response time is 70 ms, which exceeds the maximum allowed response time of 50 ms. In an autonomous vehicle, delayed detection can be dangerous because the vehicle must react quickly to surrounding objects and changes in the environment. Therefore, Method B fails the response-time constraint and cannot be selected.

Method C has an accuracy of 91% and a response time of 45 ms. Its accuracy is greater than 90%, satisfying the safety constraint. Its response time is also below 50 ms, satisfying the response-time requirement. Therefore, Method C satisfies both constraints simultaneously.

The logical comparison is:

- Method A → 88% accuracy , 30 ms response → Fails accuracy
- Method B → 93% accuracy , 70 ms response → Fails response time
- Method C → 91% accuracy , 45 ms response → Satisfies both

Although Method B has a higher accuracy than Method C, its 70 ms response time makes it unsuitable for the specified autonomous-vehicle conditions. Similarly, Method A is faster but does not provide the required level of safety.

Therefore, Method C should be selected. It provides the required accuracy while also responding within the specified 50 ms limit. This is a constraint-based decision because the method is selected only after checking every required condition. Since autonomous vehicles require both safe detection and rapid response, Method C gives the best feasible solution among the available methods.

3. Motion Analysis Decision Problem

A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic

Results:

Optical Flow: accurate but noisy in dense scenes

Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

Ans:

The traffic system must accurately detect motion direction and remain robust in **dense traffic conditions**. The two available approaches are Optical Flow and Motion Estimation. Each method has a different strength, so the decision should be based on the specific scenario rather than selecting a method solely because of one advantage.

Optical Flow provides accurate motion-direction information. This is an important advantage because the traffic system needs to determine how vehicles and other objects are moving. However, the given results state that Optical Flow becomes **noisy in dense scenes**. In high-density traffic, many vehicles may be moving close to one another, producing overlapping motion information. Noise can make the estimated motion vectors less reliable and can cause confusion when distinguishing individual vehicle movements.

Motion Estimation, on the other hand, is described as **stable but less detailed**. Stability is particularly useful in dense traffic because the system needs consistent motion information even when many objects are present simultaneously. Although Motion Estimation may not provide as much detailed motion information as Optical Flow, its stability can make it more reliable under crowded conditions.

The scenario is therefore important. In a simple traffic scene with only a few vehicles, Optical Flow could be useful because its accurate and detailed motion information may provide better direction detection. However, the requirement specifically states that the system must be **robust in dense traffic**. In this situation, noise becomes a major disadvantage. A highly detailed but noisy result may be less useful than a slightly less detailed but stable result.

Motion Estimation therefore provides a better compromise for the specified scenario. Its stable output can reduce the effect of interference caused by multiple nearby vehicles. The loss of some fine detail is an acceptable trade-off if the main objective is reliable motion analysis in crowded traffic.

The comparison can be summarized as:

- Optical Flow → accurate and detailed , but noisy in dense traffic
- Motion Estimation → less detailed, but stable in dense traffic

Therefore, **Motion Estimation should be selected for high-density traffic conditions**. The decision is based on scenario-specific requirements. Since robustness in dense traffic is explicitly required, stability is more important than maximum detail. Optical Flow may still be useful in less crowded environments or when detailed motion information is the primary objective.

Thus, for the given traffic scenario, Motion Estimation provides the more dependable solution because it prioritizes stable motion information where noise and object interactions are likely to be high.

4. Background Modeling Evaluation

A GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:

System must maintain $\geq 80\%$ accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

Ans: The GMM-based background-modeling system has to satisfy a strict requirement: it must maintain an accuracy of **at least 80% across all scenarios**. The system has been evaluated under static and dynamic conditions, producing different accuracy values.

In a **static scene**, the GMM system achieves **90% accuracy**. Since 90% is greater than the required minimum of 80%, GMM satisfies the requirement in this scenario. Static backgrounds are generally easier to model because the background remains relatively stable, allowing the Gaussian mixture model to represent the background effectively.

However, in a **dynamic scene**, the accuracy falls to only **65%**. This is below the required 80% threshold. Therefore, GMM fails the requirement under dynamic conditions. The difference between the required accuracy and the observed accuracy is **15 percentage points**. This indicates that the method loses significant performance when the background contains changing or moving elements.

The logical evaluation is:

- Static scene $\rightarrow 90\% \geq 80\% \rightarrow$ **Pass**
- Dynamic scene $\rightarrow 65\% < 80\% \rightarrow$ **Fail**
- Requirement $\rightarrow \geq 80\%$ **across all scenarios**
- Overall decision $\rightarrow$ **GMM alone does not satisfy the requirement**

The important point is that satisfying the requirement in only one scenario is not enough. The system must maintain at least 80% accuracy **across all scenarios**. Therefore, even though GMM performs well in static environments, its 65% accuracy in dynamic environments makes it unsuitable as a standalone solution.

A possible decision is to improve GMM or combine it with another technique. Since the major weakness occurs in dynamic scenes, the system could use adaptive background modeling so that it can respond to changes in the environment. Additional motion information or object-detection techniques could also be used to distinguish actual foreground objects from dynamic background elements.

A hybrid approach would allow GMM to continue handling static-background situations while another method improves detection when the background changes. The goal would be to increase the dynamic-scene accuracy from 65% to at least 80%.

Therefore, **GMM alone should not be selected** for the required system. It passes the static-scene condition but fails the overall requirement because it cannot maintain 80% accuracy in dynamic scenes. The recommended decision is to use an **improved or hybrid background-modeling approach** that specifically addresses the weaknesses of GMM in dynamic environments.

5. Depth Estimation Strategy

Constraints:

- Only one camera available
- Moderate depth accuracy acceptable

Options:

Stereo Vision: high accuracy, needs two cameras

Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

Ans:

The depth-estimation system has two main constraints. First, **only one camera is available**. Second, **moderate depth accuracy is acceptable**. Therefore, the selected technique must work with a single camera and should provide sufficient, rather than necessarily maximum, depth accuracy.

Stereo Vision generally uses two cameras to obtain depth information by comparing the views captured from different positions. It can provide high depth accuracy, which is a major advantage. However, the given constraint states that only **one camera** is available. Therefore, Stereo Vision cannot satisfy the hardware constraint. Even though its accuracy is high, it is not feasible for this particular system because the required second camera is unavailable.

Structure from Motion (SfM) provides moderate depth accuracy and can operate using a **single camera** by analyzing changes across multiple frames as the camera or scene moves. This directly satisfies the stated hardware constraint. Since the requirement says that moderate depth accuracy is acceptable, the moderate accuracy provided by SfM is sufficient for the application.

The logical comparison is:

- Stereo Vision → High accuracy , but requires two cameras
- Structure from Motion → Moderate accuracy , single camera
- Available hardware → Only one camera
- Required accuracy → Moderate accuracy is acceptable

Therefore, Stereo Vision fails the feasibility constraint even though it offers higher accuracy. Structure from Motion satisfies both the hardware and accuracy requirements.

The decision demonstrates why the highest-accuracy technique is not always the best choice. In this case, the availability of hardware is a strict constraint. Using Stereo Vision would require an additional camera, increasing hardware requirements and potentially increasing system complexity and cost. Since moderate accuracy is already acceptable, there is no need to select a method that requires unavailable hardware simply to achieve higher accuracy.

Structure from Motion is therefore the more practical solution. It can use a single camera and provide the required level of depth information. Its moderate accuracy represents an acceptable trade-off because the problem explicitly states that very high depth accuracy is not necessary.

Thus, **Structure from Motion should be selected**. It is feasible with the available single-camera setup and satisfies the application's acceptable accuracy requirement. Stereo Vision would only become a suitable option if a second camera were available and higher depth accuracy were required.